

Ethan Palosh

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EDUCATION

University of Southern California, Viterbi School of Engineering

Los Angeles, California

Bachelor of Science in Computer Engineering - May 2027

GPA: 3.62

Honors: Presidential scholarship recipient (merit aid totaling \$240k), Viterbi Dean's list scholar

Relevant Coursework: Data Structures and Algorithms, Discrete Mathematics, Object-Oriented Design, C++, Embedded Systems, Linear Algebra, Calculus 3, Probability Theory, Artificial Intelligence, Java, SQL, HTML Web Design, Software Engineering Methods

PROFESSIONAL EXPERIENCE

Software Engineering Intern

May 2024 - August 2024

ForeFlight

Austin, Texas

- Designed & implemented a scalable validation system covering 100% of user data infrastructure. *Software Design*
- Implemented a new asynchronous networking framework to support OAuth integration. *Objective-C, Swift*
- Established model UI implementation and data access layer for 100+ developers using sync team's API. *UIKit*
- Integrated version control on thousands of TeamCity automated build configs. Incorporated a PR workflow. *Kotlin*
- Developed RESTful server code using Java and SQL to develop feature updates for several other teams.

Electronics Engineer, Software Engineer, Driver

August 2023 - Present

USC Racing (Formula SAE)

Los Angeles, California

- Responsible for all software, electronic components, and wiring on car (1000+ components) w/ electronics subteam.
- Configured a new ECU system and constructed a custom dashboard system in C++ utilizing CAN networking.
- Building, testing, and documenting hundreds of components on the car's wiring harness in RapidHarness CAD.
- As driver:* Comprehensive knowledge of car and mental agility to perform in high-stakes environments.

Research Intern

May 2022 - August 2022

NASA - National Aeronautics and Space Administration

Austin, Texas

- Analyzed millions of real-world satellite data points using JavaScript and python to forecast climate disasters.
- Trained AI/machine learning models to classify cropland based on 20+ individual satellite NDVI datasets.
- [Published research:](#) "Visualizing Two Decades of Climate Disaster Impacts...via Satellite Remote Sensing..."

Software Developer, Technician

March 2022 - May 2023

FIRST Robotics, FRC 2468 Team Appreciate

Austin, Texas

- Responsible for all autonomous software and sensors on robot. **Ranked 3rd in world for autonomous control.**
- Utilized sensor fusion localization, computer vision, and object tracking for 100% of autonomous control period.
- Designed a unique autonomous path that accurately traverses variable terrain using computer vision.
- Served in community STEM outreach efforts - FRC 2468 earned top 6 in the world (finalists) for Impact Award.

PROJECTS

Full Stack Developer, Designer

May 2024 - Present

"Detailer" - [View on the iOS App Store](#)

- Scalable from-scratch iOS app** developed entirely in Swift. Complete w/ authentication, database, and remote config.
- Implemented user authentication, messaging, Google cloud functions for push notifications, RESTful API using JSON.
- Network calls for backend database through Firebase, OpenAI API, remote configured code, a robust front-end and UI.
- 160+ hours of development, 6000+ lines of swift code.. <https://thedetailerapp.com/>

SKILLS

- C++** (Both Embedded Systems & Software Engineering | Coursework)
- Swift + Xcode** (Full stack for iOS: Backend Interfacing Code, UIKit, SwiftU)
- Autonomous Control** (Computer Vision, Control Theory, Sensor Fusion)
- Python** (OpenCV/Matplotlib/Pandas)
- Java** (University Coursework)
- NI Labview** (For Robotics)

OTHER AWARDS/INTERESTS

Eagle Scout | Sports: Soccer, Pickleball, Tennis, Skiing/Snowboarding, Scuba | Motorsports/F1 | Photography